

2012-2013 – Master 2 – Macro I

Lecture 8 : Empirical studies of convergence

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Disclaimer

These are the slides I am using in class. They are not self-contained, do not always constitute original material and do contain some “cut and paste” pieces from various sources that I am not always explicitly referring to (not on purpose but because it takes time). Therefore, they are not intended to be used outside of the course or to be distributed. Thank you for signaling me typos or mistakes at franck.portier@TSE-fr.eu.

1. Facts

- ▶ The gap between rich and poor countries is large
- ▶ It is rather persistent
- ▶ Standard neo-classical models would predict rather rapid convergence

1. Facts

Productivity growth seems to accelerate over time

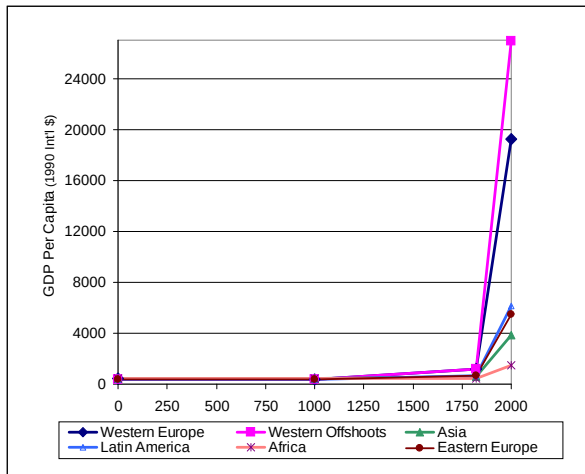
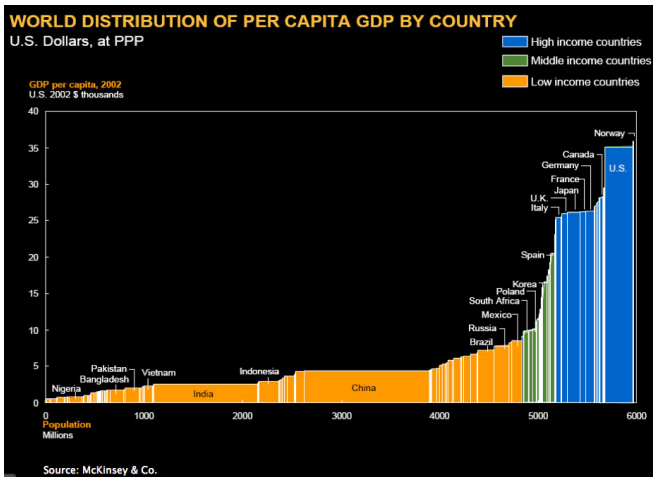


Figure 1: The Evolution of Regional Income Per Capita, 1-2000 CE
(Source: Maddison, 2003)

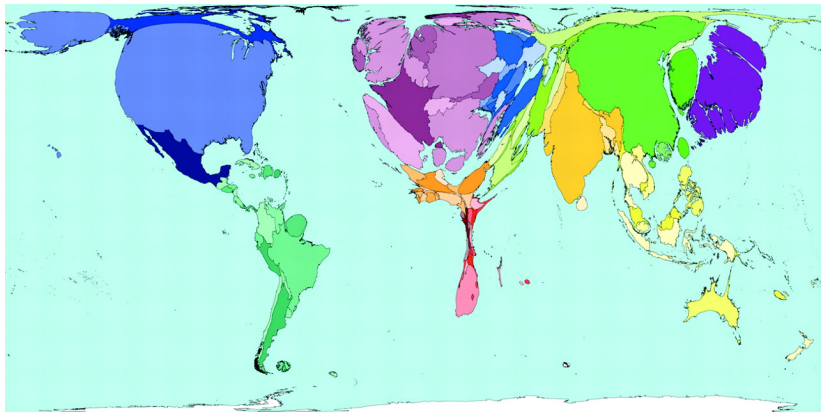
1. Facts

The gap between rich and poor countries is huge



1. Facts

The world's countries shaped with area in proportion to the gross domestic product adjusted for purchasing power parity (2002 data)



1. Facts

Can we explain it ?

- ▶ Let's take the Solow model (constant and exogenous saving rate s)
- ▶ $Y_t = A_t K_t^\alpha$
- ▶ $\dot{K}_t = sY_t - \delta K_t$
- ▶ Along a BGP : $\frac{K_t}{Y_t} = \frac{s}{g+\delta}$
- ▶ $K_t = \left(\frac{A_t s}{g+\delta} \right)^{\frac{1}{1-\alpha}} \Rightarrow Y_t = A_t^{\frac{1}{1-\alpha}} \left(\frac{s}{g+\delta} \right)^{\frac{\alpha}{1-\alpha}}$
- ▶ $\frac{\Delta Y}{Y} = \frac{\alpha}{1-\alpha} \frac{\Delta s}{s}$

1. Facts

It does not add up

- ▶ Common estimates suggest $\alpha = 1/3$
- ▶ Therefore, to have a 10-fold difference in GDP, we need a 20-fold difference in savings rate (2 % vs. 40 %)
- ▶ Could be explained if technology differed a lot across countries, but unlikely if technology is transferable
- ▶ What is the explanation ?

1. Facts

If α were greater?

- ▶ Growth would take more time to fall to zero
- ▶ Convergence would be slower (The speed of convergence is the coefficient of gdp growth on (local) initial log gdp)
- ▶ From Lecture 7, we know that the speed of convergence is $v = (g + \delta)(1 - \alpha)$
- ▶ Income differences between countries would be magnified

1. Facts

An extreme case : $\alpha = 1$, $g = 0$

- ▶ This is the *AK* model
- ▶ The speed of convergence goes to zero
- ▶ The convergence path becomes a balanced growth path at a constant rate
- ▶ MPK is no longer falling. Therefore, capital accumulation can sustain long-run growth
- ▶ The growth rate is now endogenous and depends on preferences (in a Ramsey-type model)

2. Convergence

Convergence in neo-classical models

- ▶ Neo-Classical models : each country converges to its own steady state
- ▶ All own steady states grow at the same rate
- ▶ But the level depend on policies, savings rates, etc
- ▶ Therefore, similar countries converge to same GDP per capita

2. Convergence

Convergence in endogenous growth models

- ▶ A country that starts poorer never closes the gap
- ▶ Therefore, no convergence in income levels
- ▶ This because MPK is no higher for the laggard
- ▶ Furthermore, differences in policies affect the long-run growth rate

2. Convergence

Looking at convergence allows us to :

- ▶ Test the relevance of endogenous growth models
- ▶ Assess the magnitude of the returns to accumulable factors
- ▶ Recall : $v = (g + \delta)(1 - \alpha)$

2. Convergence

Two approaches

- ▶ Barro and Sala-i-Martin : take a data set of similar economic units and look at convergence between them in per capita GDP
- ▶ Mankiw-Romer-Weil : take a cross-country regression of growth rates on initial income controlling for own long-run steady state

3. Barro and Sala-i-Martin

Estimated equation

- ▶ They use a data-base of U.S. states over a long-run period
- ▶ They estimate the equivalent of our local speed of convergence regression

$$\frac{1}{T} \times \log \left(\frac{y_{i,t_0+T}}{y_{i,t_0}} \right) = B - \left(\frac{1 - e^{-\beta T}}{T} \right) \times \log(y_{i,t_0}) + u_{i,t_0,t_0+T}$$

3. Barro and Sala-i-Martin

Results

CROSS-STATE REGRESSIONS FOR PERSONAL INCOME

Sample	$\hat{\beta}$	Sectoral Composition (s_{it})	R^2	$\hat{\sigma}$
1. 1880–1988	.0175 (.0046)	...	.92	.0014
2. 1880–1900	.0224 (.0040)	...	.62	.0054
3. 1900–1920	.0209 (.0063)	...	.67	.0062
4. 1920–30	-.0122 (.0074)	...	.43	.0111
5. 1930–40	.0127 (.0051)	...	.36	.0075
6. 1940–50	.0373 (.0053)	...	.86	.0057
7. 1950–60	.0202 (.0052)	...	.49	.0048
8. 1960–70	.0135 (.0043)	...	.68	.0037
9. 1970–80	.0119 (.0069)	...	.36	.0056
10. 1980–88	-.0005 (.0114)	...	.51	.0103

2. Convergence

Results (continued)

11. Nine periods, β restricted*	.0189 (.0019)	...	...	...
12. 1880–1900	.0268 (.0048)	-.0161 (.0079)	.65	.0053
13. 1900–1920	.0269 (.0075)	-.0214 (.0094)	.71	.0060
14. 1920–30	.0218 (.0112)	-.0936 (.0175)	.64	.0089
15. 1930–40	.0141 (.0048)	2.43 (.81)	.46	.0070
16. 1940–50	.0362 (.0055)	-.40 (.57)	.87	.0057
17. 1950–60	.0313 (.0055)	.42 (.09)	.65	.0041
18. 1960–70	.0194 (.0052)	.55 (.25)	.71	.0036
19. 1970–80	.0139 (.0076)	.25 (.37)	.36	.0056
20. 1980–88	.0196 (.0106)	1.35 (.22)	.73	.0077
21. Nine periods, β restricted*	.0249 (.0021)	individual	...	...
22. 1840–80	.0254 (.0067)	...	.91	.0030

3. Barro and Sala-i-Martin

Results (continued)

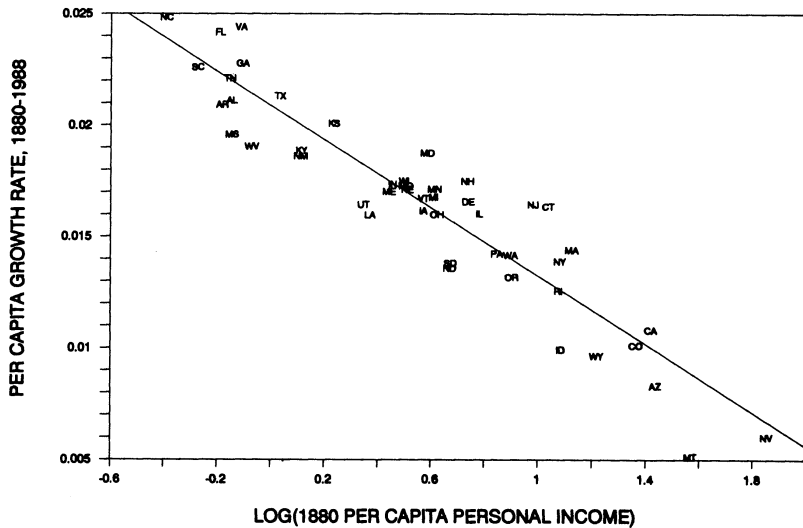


FIG. 1.—Growth rate from 1880 to 1988 vs. 1880 per capita income

3. Barro and Sala-i-Martin

The BSM Universal Law of Convergence

The speed of convergence is 2 % per year

3. Barro and Sala-i-Martin

Solow model prediction

- ▶ What do we get in a Solow model?
- ▶ The Solow model predicts a speed of convergence ν equal to $(\delta + g)(1 - \alpha)$
- ▶ A reasonable calibration is $\delta = 0.06$, $g = 0.02$, $\alpha = 0.3$
- ▶ This gives $\nu = 5.6$ % per year

3. Barro and Sala-i-Martin

How universal is the law ?

TABLE 3
COMPARISON OF REGRESSIONS ACROSS COUNTRIES AND U.S. STATES

Sample	$\hat{\beta}$	Additional Variables	R^2	$\hat{\sigma}$
1. 98 countries, 1960–85	–.0037 (.0018)	no	.04	.0183
2. 98 countries, 1960–85	.0184 (.0045)	yes	.52	.0133
3. 20 OECD countries, 1960–85	.0095 (.0028)	no	.45	.0051
4. 20 OECD countries, 1960–85	.0203 (.0068)	yes	.69	.0046
5. 48 U.S. states, 1963–86	.0218 (.0053)	no	.38	.0040
6. 48 U.S. states, 1963–86	.0236 (.0013)	yes	.61	.0033

3. Barro and Sala-i-Martin

How universal is the law ? (continued)

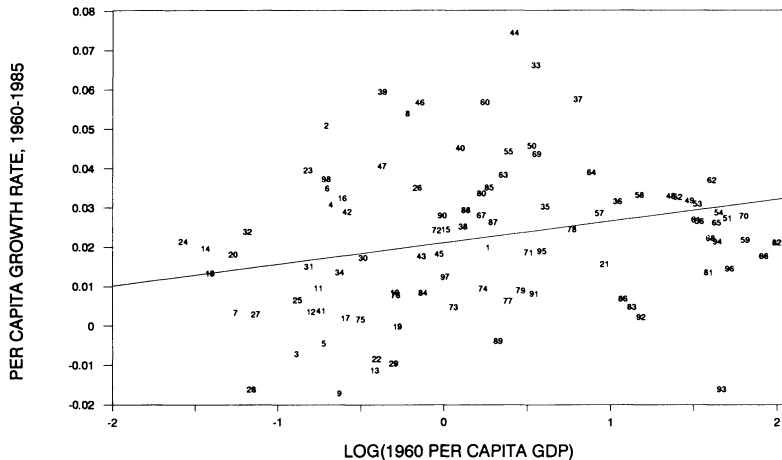


FIG. 4.—Growth rate from 1960 to 1985 vs. 1960 per capita GDP, sample of 98 countries (listed in App. B).

3. Barro and Sala-i-Martin

Findings

- ▶ The more similar the countries, the more it holds unconditionally
- ▶ The less similar the countries, the more likely we find divergence
- ▶ But the law is restored if controls are added, controlling for own steady state

3. Barro and Sala-i-Martin

Implication for how to eradicate poverty?

1. Adopt the policies and institutions of advanced countries
2. And then *wait* for convergence ...
 - ▶ How long? Suppose I am 10 times poorer than the US. How long does it take to be 2 times poorer?
 - ▶ We need to solve $\frac{d}{dt} \log \left(\frac{Y(t)}{Y_{US}(t)} \right) = -\beta \log \left(\frac{Y(t)}{Y_{US}(t)} \right)$
 - ▶ which implies
$$\log Y(t) - \log Y_{US}(t) = (\log(Y(0)) - \log(Y_{US}(0))) \times e^{-\beta t}$$
 - ▶ and therefore $t = -\frac{1}{\beta} \log \left(\frac{\log \rho_1}{\log \rho_0} \right)$

3. Barro and Sala-i-Martin

Wait... For how long?

- ▶ With $v = 0.02$, $\rho_0 = 0.1$, $\rho_1 = 0.5$, we obtain $t = 60$ years!
- ▶ With $v = 0.056$, get instead $t = 21$ years.
- ▶ We want to understand why the speed of convergence is so low
- ▶ Can policy increase the speed of convergence?
- ▶ In principle, the speed of convergence only depends on the deep technological parameters
- ▶ That it is low tells us that the technology is not what we thought it was
- ▶ But it does not tell us if and how we can increase v
- ▶ A gloomy perspective...

4. Mankiw, Romer and Weil

Approach

- ▶ National accounts suggest that the elasticity to capital α is 0.3
- ▶ Observed speeds of convergence suggest
$$\alpha = 1 - v/(g + \delta) = 1 - 0.02/0.08 = 0.75$$
- ▶ To reconcile these two facts, MRW introduce another form of capital : Human capital

4. Mankiw, Romer and Weil

The Augmented Solow model

- ▶ $Y(t) = K(t)^\alpha H(t)^\beta (A(t)L(t))^{1-\alpha-\beta}$
- ▶ $\dot{K}(t) = s_k Y(t) - \delta K(t)$
- ▶ $\dot{H}(t) = s_h Y(t) - \delta H(t)$
- ▶ Denoting $x(t) = \frac{X(t)}{A(t)L(t)}$, we have
 - ▶ $\dot{k}(t) = s_k y(t) - (n + g + \delta)k(t)$
 - ▶ $\dot{h}(t) = s_h y(t) - (n + g + \delta)h(t)$

4. Mankiw, Romer and Weil

BGP

- ▶ Along a BGP, deflated variables are at a Steady State :

- ▶ $k^* = \left(\frac{s_k^{1-\beta} s_h^\beta}{n+g+\delta} \right)^{\frac{1}{1-\alpha-\beta}}$

- ▶ $h^* = \left(\frac{s_k^\alpha s_h^{1-\alpha}}{n+g+\delta} \right)^{\frac{1}{1-\alpha-\beta}}$

4. Mankiw, Romer and Weil

Explaining cross-country differences in per capita GDP :

- ▶ The preceding equations define “own” steady state
- ▶ MRW use it to see if it explains cross-country income differences :

$$\begin{aligned}\log \left(\frac{Y(t)}{L(t)} \right) &= \log A(0) + g \times t - \frac{\alpha + \beta}{1 - \alpha - \beta} \log(n + g + \delta) \\ &+ \frac{\alpha}{1 - \alpha - \beta} \log s_k + \frac{\beta}{1 - \alpha - \beta} \log s_h\end{aligned}$$

4. Mankiw, Romer and Weil

Measuring s_h

We use a proxy for the rate of human-capital accumulation (s_h) that measures approximately the percentage of the working-age population that is in secondary school. We begin with data on the fraction of the eligible population (aged 12 to 17) enrolled in secondary school, which we obtained from the UNESCO yearbook. We then multiply this enrollment rate by the fraction of the working-age population that is of school age (aged 15 to 19). This variable, which we call SCHOOL, is clearly imperfect: the age ranges in the two data series are not exactly the same, the variable does not include the input of teachers, and it completely ignores primary and higher education. Yet if SCHOOL is proportional to s_h , then we can use it to estimate equation (11); the factor of proportionality will affect only the constant term.¹⁰

4. Mankiw, Romer and Weil

Results

TABLE II
ESTIMATION OF THE AUGMENTED SOLOW MODEL

Dependent variable: log GDP per working-age person in 1985			
Sample:	Non-oil	Intermediate	OECD
Observations:	98	75	22
CONSTANT	6.89 (1.17)	7.81 (1.19)	8.63 (2.19)
$\ln(I/GDP)$	0.69 (0.13)	0.70 (0.15)	0.28 (0.39)
$\ln(n + g + \delta)$	-1.73 (0.41)	-1.50 (0.40)	-1.07 (0.75)
$\ln(SCHOOL)$	0.66 (0.07)	0.73 (0.10)	0.76 (0.29)
$\bar{R}^2$	0.78	0.77	0.24
<i>s.e.e.</i>	0.51	0.45	0.33
Restricted regression:			
CONSTANT	7.86 (0.14)	7.97 (0.15)	8.71 (0.47)
$\ln(I/GDP) - \ln(n + g + \delta)$	0.73 (0.12)	0.71 (0.14)	0.29 (0.33)
$\ln(SCHOOL) - \ln(n + g + \delta)$	0.67 (0.07)	0.74 (0.09)	0.76 (0.28)
$\bar{R}^2$	0.78	0.77	0.28
<i>s.e.e.</i>	0.51	0.45	0.32
Test of restriction:			
<i>p</i> -value	0.41	0.89	0.97
Implied α	0.31 (0.04)	0.29 (0.05)	0.14 (0.15)
Implied β	0.28 (0.03)	0.30 (0.04)	0.37 (0.12)

Note. Standard errors are in parentheses. The investment and population growth rates are averages for the period 1960–1985. $(g + \delta)$ is assumed to be 0.05. SCHOOL is the average percentage of the working-age population in secondary school for the period 1960–1985.

4. Mankiw, Romer and Weil

What have we learned?

- ▶ We have seen previously that with $\alpha = 0.3$, it is difficult to explain country income differences
- ▶ But now what matters is $\alpha + \beta$, which acts as α .
- ▶ So with $\alpha + \beta$ large enough we can explain cross-country differences.
- ▶ A natural question is : can we also expect slow convergence?

4. Mankiw, Romer and Weil

Recomputing the speed of convergence

- ▶ Let $x = \frac{X}{X_{LR}}$
- ▶ One must have $s_k \frac{Y_{LR}}{K_{LR}} = n + g + \delta$
- ▶ Define $\hat{y} = \log y$, then $v = -\frac{\dot{\hat{y}}}{\hat{y}}$.
- ▶ $\frac{\dot{y}}{y} = \alpha \frac{\dot{k}}{k} + \beta \frac{\dot{h}}{h}$
- ▶ $\begin{aligned} \frac{\dot{k}}{k} &= \frac{\dot{K}}{K} - (n + g) = s_k \frac{Y}{K} - (\delta + n + g) \\ &= s_k \frac{Y_{LR}}{K_{LR}} (1 - \hat{y} - \hat{k}) - (\delta + n + g) = (\delta + n + g)(\hat{y} - \hat{k}) \end{aligned}$
- ▶ Similarly, $\frac{\dot{h}}{h} = (\delta + n + g)(\hat{y} - \hat{h})$
- ▶ Therefore, $\frac{\dot{y}}{y} = \alpha(\delta + n + g)(\hat{y} - \hat{k}) + \beta(\delta + n + g)(\hat{y} - \hat{h}) = -(\delta + n + g)(1 - \alpha - \beta)\hat{y}$

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Empirical strategy

- ▶ Investment rates and schooling are kept to proxy for own steady state
- ▶ Initial output is added
- ▶ Coefficient in initial output related to SOV as in BSM
- ▶ No other control variable is added in strict interpretation of Solow model

4. Mankiw, Romer and Weil

Old Solow does not work ...

TABLE IV
TESTS FOR CONDITIONAL CONVERGENCE

Dependent variable: log difference GDP per working-age person 1960–1985			
Sample:	Non-oil	Intermediate	OECD
Observations:	98	75	22
CONSTANT	1.93 (0.83)	2.23 (0.86)	2.19 (1.17)
$\ln(Y60)$	-0.141 (0.052)	-0.228 (0.057)	-0.351 (0.066)
$\ln(I/GDP)$	0.647 (0.087)	0.644 (0.104)	0.392 (0.176)
$\ln(n + g + \delta)$	-0.299 (0.304)	-0.464 (0.307)	-0.753 (0.341)
$\bar{R}^2$	0.38	0.35	0.62
<i>s.e.e.</i>	0.35	0.33	0.15
Implied λ	0.00606 (0.00182)	0.0104 (0.0019)	0.0173 (0.0019)

Note. Standard errors are in parentheses. Y60 is GDP per working-age person in 1960. The investment and population growth rates are averages for the period 1960–1985. $(g + \delta)$ is assumed to be 0.05.

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... but new does.

TABLE V
TESTS FOR CONDITIONAL CONVERGENCE

Dependent variable: log difference GDP per working-age person 1960–1985			
Sample:	Non-oil	Intermediate	OECD
Observations:	98	75	22
CONSTANT	3.04 (0.83)	3.69 (0.91)	2.81 (1.19)
$\ln(Y60)$	-0.289 (0.062)	-0.366 (0.067)	-0.398 (0.070)
$\ln(I/GDP)$	0.524 (0.087)	0.538 (0.102)	0.335 (0.174)
$\ln(n + g + \delta)$	-0.505 (0.288)	-0.551 (0.288)	-0.844 (0.334)
$\ln(SCHOOL)$	0.233 (0.060)	0.271 (0.081)	0.223 (0.144)
$\bar{R}^2$	0.46	0.43	0.65
<i>s.e.e.</i>	0.33	0.30	0.15
Implied λ	0.0137 (0.0019)	0.0182 (0.0020)	0.0203 (0.0020)

Note. Standard errors are in parentheses. Y60 is GDP per working-age person in 1960. The investment and population growth rates are averages for the period 1960–1985. $(g + \delta)$ is assumed to be 0.05. SCHOOL is the average percentage of the working-age population in secondary school for the period 1960–1985.

4. Mankiw, Romer and Weil

Does it add up?

- ▶ $\alpha = .3$
- ▶ $\beta = .3$
- ▶ $n + g + \delta = .06$
- ▶ Therefore $v = .024$

5. Summary

- ▶ The Solow model predicts too low income disparities and too quick convergence
- ▶ The AK model predicts zero convergence and widening disparities
- ▶ The Augmented Solow model does well to predict both the disparities and the speed of convergence